

IR-3 Elements

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IR-3.1 NAMES AND SYMBOLS OF ATOMS

The origins of the names of some chemical elements, for example antimony, are lost in antiquity. Other elements recognized (or discovered) during the past three centuries were named according to various associations of origin, physical or chemical properties, *etc.*, and more recently to commemorate the names of eminent scientists.

In the past, some elements were given two names because two groups claimed to have discovered them. To avoid such confusion it was decided in 1947 that after the existence of a new element had been proved beyond reasonable doubt, discoverers had the right to *suggest* a name to IUPAC, but that only the Commission on Nomenclature of Inorganic Chemistry (CNIC) could make a recommendation to the IUPAC Council to make the final decision.

Under the present procedure,¹ claims of the discovery of a new element are first investigated by a joint IUPAC-IUPAP (International Union of Pure and Applied Physics) committee which then assigns priority. The acknowledged discoverers are then invited to *suggest* a name to the Division of Inorganic Chemistry, which then makes a formal recommendation to the IUPAC Council. It is emphasized that the name for any new element resulting from the above process, once approved by Council, is not intended to carry any implication regarding priority of discovery. The same is understood to apply to the names approved by IUPAC in the past, whatever the prehistory.

The IUPAC-approved names of the atoms of atomic numbers 1-111 for use in the English language are listed in alphabetical order in Table I*. It is obviously desirable that the names used in any language resemble these names as closely as possible, but it is recognized that for elements named in the past there are often well-established and very different names in other languages. In the footnotes of Table I, certain names are cited which are not used now in English, but which either provide the basis of the atomic symbol, or the basis of certain affixes used in nomenclature.

For use in chemical formulae, each atom is represented by a unique symbol in upright type as shown in Table I. In addition, the symbols D and T may be used for the hydrogen isotopes of mass numbers two and three, respectively (see Section IR-3.3.2).

IR-3.1.1 Systematic nomenclature and symbols for new elements

Newly discovered elements may be referred to in the scientific literature but until they have received permanent names and symbols from IUPAC, temporary designators are required. Such elements may be referred to by their atomic numbers, as in 'element 120' for example, but IUPAC has approved a systematic nomenclature and series of three-letter symbols (see Table II).²

The name is derived directly from the atomic number of the element using the following numerical roots:

0 = nil	3 = tri	6 = hex	9 = enn
1 = un	4 = quad	7 = sept	
2 = bi	5 = pent	8 = oct	

The roots are put together in the order of the digits which make up the atomic number and terminated by 'ium' to spell out the name. The final 'n' of 'enn' is elided when it occurs before 'nil', and the final 'i' of 'bi' and of 'tri' when it occurs before 'ium'.

The symbol for the element is composed of the initial letters of the numerical roots which make up the name.

Example:

1. element 113 = ununtrium, symbol Uut

IR-3.2 INDICATION OF MASS, CHARGE AND ATOMIC NUMBER USING INDEXES (SUBSCRIPTS AND SUPERSSCRIPTS)

The mass, charge and atomic number of a nuclide are indicated by means of three indexes (subscripts and superscripts) placed around the symbol. The positions are occupied as follows:

left upper index	mass number
left lower index	atomic number
right upper index	charge

A charge placed on an atom of symbol A is indicated as A^{n+} or A^{n-} , not as A^{+n} or A^{-n} .

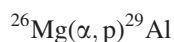
* Tables numbered with a Roman numeral are collected together at the end of this book.

The right lower position of an atomic symbol is reserved for an index (subscript) indicating the number of such atoms in a formula. For example, S_8 is the formula of a molecule containing eight sulfur atoms (see Section IR-3.4). For formalisms when oxidation states or charges are also shown, see Section IR-4.6.1.

Example:

1. ${}^{32}_{16}S^{2+}$ represents a doubly ionized sulfur atom of atomic number 16 and mass number 32.

The nuclear reaction between ${}^{26}_{12}Mg$ and 4_2He nuclei to yield ${}^{29}_{13}Al$ and 1_1H nuclei is written as follows³:



For the use of atomic symbols to indicate isotopic modification in chemical formulae and the nomenclature of isotopically modified compounds see Section IR-4.5 and Chapter II-2 of Ref. 4 respectively.

IR-3.3 ISOTOPES

IR-3.3.1 **Isotopes of an element**

The isotopes of an element all bear the same name (but see Section IR-3.3.2) and are designated by mass numbers (see Section IR-3.2). For example, the atom of atomic number 8 and mass number 18 is named oxygen-18 and has the symbol ${}^{18}O$.

IR-3.3.2 **Isotopes of hydrogen**

Hydrogen is an exception to the rule in Section IR-3.3.1 in that the three isotopes 1H , 2H and 3H can have the alternative names protium, deuterium and tritium, respectively. The symbols D and T may be used for deuterium and tritium but 2H and 3H are preferred because D and T can disturb the alphabetical ordering in formulae (see Section IR-4.5). The combination of a muon and an electron behaves like a light isotope of hydrogen and is named muonium, symbol Mu.⁵

These names give rise to the names proton, deuteron, triton and muon for the cations ${}^1H^+$, ${}^2H^+$, ${}^3H^+$ and Mu^+ , respectively. Because the name proton is often used in contradictory senses, *i.e.* for isotopically pure ${}^1H^+$ ions on the one hand, and for the naturally occurring undifferentiated isotope mixture on the other, it is recommended that the undifferentiated mixture be designated generally by the name hydron, derived from hydrogen.

IR-3.4 ELEMENTS (or elementary substances)

IR-3.4.1 **Name of an element of indefinite molecular formula or structure**

A sample of an element that has an undefined formula, or is a mixture of allotropes (see Sections IR-3.4.2 to IR-3.4.5), bears the same name as the atom.

IR-3.4.2 Allotropes (allotropic modifications) of elements

Allotropic modifications of an element bear the name of the atom from which they are derived, together with a descriptor to specify the modification. Common descriptors are Greek letters (α , β , γ , *etc.*), colours and, where appropriate, mineral names (*e.g.* graphite and diamond for the well known forms of carbon). Such names should be regarded as provisional, to be used only until structures have been established, after which a rational system based on molecular formula (see Section IR-3.4.3) or crystal structure (see Section IR-3.4.4) is recommended. Common names will continue to be used for amorphous modifications of an element and for those which are mixtures of closely related structures in their commonly occurring forms (such as graphite) or have an ill-defined disordered structure (such as red phosphorus) (see Section IR-3.4.5).

IR-3.4.3 Names of allotropes of definite molecular formula

Systematic names are based on the number of atoms in the molecule, indicated by a multiplicative prefix from Table IV. The prefix 'mono' is only used when the element does not normally occur in a monoatomic state. If the number is large and unknown, as in long chains or large rings, the prefix 'poly' may be used. Where necessary, appropriate prefixes (Table V) may be used to indicate structure. When it is desired to specify a particular polymorph of an element with a defined structure (such as the α -, β - or γ -forms of S_8) the method of Section IR-3.4.4 should be used (see Examples 13–15 in Section IR-3.4.4).

Examples:

	<i>Formula</i>	<i>Systematic name</i>	<i>Acceptable alternative name</i>
1.	Ar	argon	
2.	H	monohydrogen	
3.	N	mononitrogen	
4.	N ₂	dinitrogen	
5.	N ₃ [•]	trinitrogen(•)	
6.	O ₂	dioxygen	oxygen
7.	O ₃	trioxygen	ozone
8.	P ₄	tetraphosphorus	white phosphorus
9.	S ₆	hexasulfur	ϵ -sulfur
10.	S ₈	<i>cyclo</i> -octasulfur	α -sulfur, β -sulfur, γ -sulfur
11.	S _{<i>n</i>}	polysulfur	μ -sulfur (or plastic sulfur)
12.	C ₆₀	hexacontacarbon	[60]fullerene

In Example 12, the name [60]fullerene is to be regarded as an acceptable non-systematic name for a particular C₆₀ structure. For more details see Section P-27 of Ref. 6.

IR-3.4.4 Crystalline allotropic modifications of an element

Crystalline allotropic modifications are polymorphs of the elements. Each can be named by adding the Pearson symbol (see Section IR-11.5.2)⁷ in parentheses after the name of the

atom. This symbol defines the structure of the allotrope in terms of its Bravais lattice (crystal class and type of unit cell, see Table IR-3.1) and the number of atoms in its unit cell. Thus, iron(*cF4*) is the allotropic modification of iron (γ -iron) with a cubic (*c*), all-face-centred (*F*) lattice containing four atoms of iron in the unit cell.

Table IR-3.1 *Pearson symbols used for the fourteen Bravais lattices*

<i>System</i>	<i>Lattice symbol^a</i>	<i>Pearson symbol</i>
Triclinic	<i>P</i>	<i>aP</i>
Monoclinic	<i>P</i>	<i>mP</i>
	<i>S^b</i>	<i>mS</i>
Orthorhombic	<i>P</i>	<i>oP</i>
	<i>S</i>	<i>oS</i>
	<i>F</i>	<i>oF</i>
	<i>I</i>	<i>oI</i>
Tetragonal	<i>P</i>	<i>tP</i>
	<i>I</i>	<i>tI</i>
Hexagonal (and trigonal <i>P</i>)	<i>P</i>	<i>hP</i>
Rhombohedral	<i>R</i>	<i>hR</i>
Cubic	<i>P</i>	<i>cP</i>
	<i>F</i>	<i>cF</i>
	<i>I</i>	<i>cI</i>

^a *P*, *S*, *F*, *I* and *R* are primitive, side-face-centred, all-face-centred, body-centred and rhombohedral lattices, respectively. The letter *C* was formerly used in place of *S*.

^b Second setting, y -axis unique.

Examples:

	<i>Symbol</i>	<i>Systematic name</i>	<i>Acceptable alternative name</i>
1.	P_n	phosphorus(<i>oS8</i>)	black phosphorus
2.	C_n	carbon(<i>cF8</i>)	diamond
3.	C_n	carbon(<i>hP4</i>)	graphite (common form)
4.	C_n	carbon(<i>hR6</i>)	graphite (less common form)
5.	Fe_n	iron(<i>cI2</i>)	α -iron
6.	Fe_n	iron(<i>cF4</i>)	γ -iron
7.	Sn_n	tin(<i>cF8</i>)	α - or grey tin
8.	Sn_n	tin(<i>tI4</i>)	β - or white tin
9.	Mn_n	manganese(<i>cI58</i>)	α -manganese
10.	Mn_n	manganese(<i>cP20</i>)	β -manganese
11.	Mn_n	manganese(<i>cF4</i>)	γ -manganese
12.	Mn_n	manganese(<i>cI2</i>)	δ -manganese

13.	S ₈	sulfur(<i>oF</i> 128)	α-sulfur
14.	S ₈	sulfur(<i>mP</i> 48)	β-sulfur
15.	S ₈	sulfur(<i>mP</i> 32)	γ-sulfur

In a few cases, the Pearson symbol fails to differentiate between two crystalline allotropes of the element. In such an event the space group is added to the parentheses. If this still fails to distinguish the allotropes, the characteristically different lattice parameters will have to be cited. An alternative notation involving compound type may also be useful (see Section IR-4.2.5 and Chapter IR-11).

IR-3.4.5 Solid amorphous modifications and commonly recognized allotropes of indefinite structure

Solid amorphous modifications and commonly recognized allotropes of indefinite structure are distinguished by customary descriptors such as a Greek letter, names based on physical properties, or mineral names.

Examples:

1. C_n vitreous carbon
2. C_n graphitic carbon (carbon in the form of graphite, irrespective of structural defects)
3. P_n red phosphorus [a disordered structure containing parts of phosphorus(*oS*8) and parts of tetrphosphorus]
4. As_n amorphous arsenic

IR-3.5 ELEMENTS IN THE PERIODIC TABLE

The groups of elements in the periodic table (see inside front cover) are numbered from 1 to 18. The elements (except hydrogen) of groups 1, 2 and 13–18 are designated as main group elements and, except in group 18, the first two elements of each main group are termed typical elements. Optionally, the letters s, p, d and f may be used to distinguish different blocks of elements. For example, the elements of groups 3–12 are the d-block elements. These elements are also commonly referred to as the transition elements, though the elements of group 12 are not always included; the f-block elements are sometimes referred to as the inner transition elements. If appropriate for a particular purpose, the various groups may be named from the first element in each, for example elements of the boron group (B, Al, Ga, In, Tl), elements of the titanium group (Ti, Zr, Hf, Rf), *etc.*

The following collective names for like elements are IUPAC-approved: alkali metals (Li, Na, K, Rb, Cs, Fr), alkaline earth metals (Be, Mg, Ca, Sr, Ba, Ra), pnictogens⁸ (N, P, As, Sb, Bi), chalcogens (O, S, Se, Te, Po), halogens (F, Cl, Br, I, At), noble gases (He, Ne, Ar, Kr, Xe, Rn), lanthanoids (La, Ce, Pr, Nd, Pm, Sm, Eu, Gd, Tb, Dy, Ho, Er, Tm, Yb, Lu), rare earth metals (Sc, Y and the lanthanoids) and actinoids (Ac, Th, Pa, U, Np, Pu, Am, Cm, Bk, Cf, Es, Fm, Md, No, Lr).

The generic terms pnictide, chalcogenide and halogenide (or halide) are commonly used in naming compounds of the pnictogens, chalcogens and halogens.

Although lanthanoid means ‘like lanthanum’ and so should not include lanthanum, lanthanum has become included by common usage. Similarly, actinoid. The ending ‘ide’ normally indicates a negative ion, and therefore lanthanoid and actinoid are preferred to lanthanide and actinide.

IR-3.6

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8. The alternative spelling ‘pnicogen’ is also used.